

Classwork – Tuples in Python

Question 1: Employee Performance Evaluation

Problem Statement

A company stores employee details in a tuple. Each employee record contains:

```
employees = (  
    ("E101", "Anuj", 92),  
    ("E102", "Rahul", 76),  
    ("E103", "Priya", 58),  
    ("E104", "Neha", 88),  
    ("E105", "Amit", 45)  
)
```

Where:

- First value = Employee ID
- Second value = Employee Name
- Third value = Performance Score

Tasks

Write a Python program to:

1. Display details of employees scoring **80 or above**.
2. Count the number of employees who need improvement (score below 60).
3. Find the employee with the highest score.
4. Create a list containing the names of all employees scoring above 75.
5. Display the performance category for each employee:
 - 90 and above → Excellent
 - 75 to 89 → Good
 - 60 to 74 → Average
 - Below 60 → Needs Improvement

Sample Output

Employees Scoring 80 or Above:

E101 Anuj 92

E104 Neha 88

Employees Needing Improvement: 2

Highest Performer:

E101 Anuj 92

High Performers:

['Anuj', 'Rahul', 'Neha']

Performance Categories:

Anuj -> Excellent

Rahul -> Good

Priya -> Needs Improvement

Neha -> Good

Amit -> Needs Improvement

Question 2: Flight Booking Analysis

Problem Statement

A flight reservation system stores passenger records as tuples:

```
bookings = (  
    ("P101", "Delhi", "Confirmed"),  
    ("P102", "Mumbai", "Waiting"),  
    ("P103", "Delhi", "Confirmed"),  
    ("P104", "Chennai", "Cancelled"),  
    ("P105", "Mumbai", "Confirmed"),  
    ("P106", "Delhi", "Waiting")  
)
```

Where:

- Passenger ID
- Destination
- Booking Status

Tasks

Write a Python program to:

1. Display all passengers whose booking status is **Confirmed**.
2. Count the number of passengers travelling to **Delhi**.
3. Count Confirmed, Waiting, and Cancelled bookings separately.
4. Create a list containing passenger IDs with Waiting status.
5. Determine which destination has the highest number of bookings.

Sample Output

Confirmed Passengers:

P101 Delhi

P103 Delhi

P105 Mumbai

Passengers Travelling to Delhi: 3

Confirmed: 3

Waiting: 2

Cancelled: 1

Waiting List:

['P102', 'P106']

Most Booked Destination:

Delhi